Practical – 1 – OS

Open | practical1.c ~/ossp_practical | Save

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    /* Remove newline */
    command[strcspn(command, "\n")] = '\0';

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        /* Child process */
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, (char *)NULL);

        /* Only reached if exec fails */
        perror("exec failed");
        exit(1);
    }
    else
    {
        /* Parent process */
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

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Practical – 1 – OS

```
Saketh@SakethVutharavalli:~$ mkdir -p ~/ossp_practical
cd ~/ossp_practical
Saketh@SakethVutharavalli:~/ossp_practical$ gedit practical1.c
Saketh@SakethVutharavalli:~/ossp_practical$ gcc practical1.c -o practical1
Saketh@SakethVutharavalli:~/ossp_practical$ ./practical1
Enter a Linux command: ls

Parent Process
Parent PID: 552
Child PID: 553

Child Process
Child PID: 553
Parent PID: 552
practical1  practical1.c
Child process completed.
Saketh@SakethVutharavalli:~/ossp_practical$ gedit practical1.c
```